

Codebook: VVIQ and TAS-20 data across five studies (aphantasia / alexithymia)

Pooled participant-level data from five independent studies, combining responses to the Vividness of Visual Imagery Questionnaire (VVIQ; Marks, 1973) and the 20-item Toronto Alexithymia Scale (TAS-20; Bagby, Parker, & Taylor, 1994). Used to examine the relationship between visual mental imagery vividness (including aphantasia and hyperphantasia) and alexithymia.

Sources

- Ale & Burns (2024) - <https://doi.org/10.31234/osf.io/kj5d3> (study = “burns”)
- Monzel et al. (2024) - <https://doi.org/10.1016/j.bionps.2024.100106> (study = “monzel”)
- Ruby (2025) - unpublished, Perrine Ruby, Centre de Recherche en Neurosciences de Lyon (study = “ruby”)
- Mas & Luminet (2025) - unpublished, Marine Mas and Olivier Luminet, Louvain-la-Neuve (study = “mas”)
- Kvamme et al. (2026) - <https://doi.org/10.1016/j.neuropsychologia.2026.109368> (study = “kvamme”)

Instrument references and copyright notice

VVIQ and TAS-20 item wording is copyrighted by the original scale authors and is not reproduced here. Item columns below are identified by their official item number and (for the TAS-20) subscale; consult the source publications above for exact wording.

- VVIQ item text: Marks, D. F. (1973). Visual imagery differences in the recall of pictures. *British Journal of Psychology*, 64(1), 17-24. <https://doi.org/10.1111/j.2044-8295.1973.tb01322.x>
- TAS-20 item text: Bagby, R. M., Parker, J. D. A., & Taylor, G. J. (1994). The twenty-item Toronto Alexithymia Scale-I. *Journal of Psychosomatic Research*, 38(1), 23-32. [https://doi.org/10.1016/0022-3999\(94\)90005-1](https://doi.org/10.1016/0022-3999(94)90005-1)

Variables

Core variables

Variable	Description	Type	Range	Levels / notes
study	Study of origin.	string		burns = Ale & Burns (2024); monzel = Monzel et al. (2024); ruby = Ruby (2025), unpublished; mas = Mas & Luminet (2025), unpublished; kvamme = Kvamme et al. (2026)
lang	Language in which the study was administered.	string		en = English; fr = French

Variable	Description	Type	Range	Levels / notes
id	Unique participant identifier, of the form “subj_____”. Unique across the whole pooled dataset (not just within study).	string		
sex	Sex assigned at birth, as reported by the participant.	string		male = Male; female = Female; other = Other Distinct from gender below. For most participants sex and gender coincide; the Burns study collected both constructs separately and a small number of participants (n = 4) reported a sex and gender that differ.
gender	Gender identity, as reported by the participant, where collected separately from sex.	string		male = Male; female = Female; other = Other
age	Participant age in years, self-reported.	integer	10-86 years	Missing for 3 participants.
vviq	Total score on the Vividness of Visual Imagery Questionnaire (VVIQ; Marks, 1973), summed across all 16 items (see vviq_q1-vviq_q16 below). Lower scores indicate weaker/absent visual imagery.	integer	16-80	

Variable	Description	Type	Range	Levels / notes
<code>tas</code>	Total score on the 20-item Toronto Alexithymia Scale (TAS-20; Bagby, Parker, & Taylor, 1994), summed across all 20 items (see <code>tas_q1-tas_q20</code> below, already reverse-scored where applicable). Higher scores indicate greater alexithymia.	integer	20-100	
<code>tas_identify</code>	Difficulty Identifying Feelings (DIF) subscale of the TAS-20. Sum of items 1, 3, 6, 7, 9, 13, 14 (7 items).	integer	7-35	
<code>tas_describe</code>	Difficulty Describing Feelings (DDF) subscale of the TAS-20. Sum of items 2, 4, 11, 12, 17 (5 items).	integer	5-25	
<code>tas_external</code>	Externally Oriented Thinking (EOT) subscale of the TAS-20. Sum of items 5, 8, 10, 15, 16, 18, 19, 20 (8 items).	integer	8-40	
<code>tas_group</code>	Categorical alexithymia status based on standard TAS-20 cutoff (total score ≥ 61 = alexithymia).	string		<code>typical_tas</code> = Total TAS-20 score 20-60 (non-alexithymic); <code>alexithymia</code> = Total TAS-20 score 61-100 (alexithymic)

Variable	Description	Type	Range	Levels / notes
vviq_group_4	Categorical VVIQ group with 4 levels, based on standard VVIQ cutoffs.	string		aphantasia = VVIQ = 16 (complete absence of visual imagery); hypophantasia = VVIQ 17-32 (reduced visual imagery); typical = VVIQ 33-74 (typical visual imagery); hyperphantasia = VVIQ 75-80 (extremely vivid visual imagery)
vviq_group_3	Categorical VVIQ group with 3 levels (typical and hyperphantasia collapsed).	string		aphantasia = VVIQ = 16; hypophantasia = VVIQ 17-32; typical = VVIQ 33-80 (typical and hyperphantasia collapsed)
vviq_group_2	Categorical VVIQ group with 2 levels (broadest split).	string		aphantasia = VVIQ 16-32 (aphantasia and hypophantasia collapsed); typical = VVIQ 33-80 (typical and hyperphantasia collapsed)

VVIQ item-level responses (vviq_q1-vviq_q16)

The VVIQ has no subscales; its 16 items are grouped into 4 scenes of 4 items each (familiar person, sunrise, shop, natural scene), each rated 1 (perfectly clear and vivid) to 5 (no image at all). Item wording is not reproduced here for copyright reasons — see Marks (1973) for the original items. **Available for the burns, mas, ruby, and kvamme studies only; missing (NA) for all 105 monzel participants**, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
vviq_q1	VVIQ item 1 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q2	VVIQ item 2 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q3	VVIQ item 3 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q4	VVIQ item 4 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q5	VVIQ item 5 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
vviq_q6	VVIQ item 6 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q7	VVIQ item 7 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q8	VVIQ item 8 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q9	VVIQ item 9 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q10	VVIQ item 10 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
vviq_q11	VVIQ item 11 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q12	VVIQ item 12 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q13	VVIQ item 13 of 16, from the scene “a country scene with trees, mountains, and a lake” (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q14	VVIQ item 14 of 16, from the scene “a country scene with trees, mountains, and a lake” (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
vviq_q15	VVIQ item 15 of 16, from the scene “a country scene with trees, mountains, and a lake” (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
vviq_q16	VVIQ item 16 of 16, from the scene “a country scene with trees, mountains, and a lake” (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

TAS-20 item-level responses (tas_q1-tas_q20)

20 items rated 1 (strongly disagree) to 5 (strongly agree), across three subscales (DIF, DDF, EOT; see core variables above). Item wording is not reproduced here for copyright reasons — see Bagby, Parker, & Taylor (1994) for the original items. **Values in these columns are already reverse-scored for the 5 reverse-keyed items (4, 5, 10, 18, 19)** — summing tas_q1-tas_q20 directly reproduces the tas total column, with no further reversal needed. Available for the burns, mas, ruby, and kvamme studies only; missing (NA) for all 105 monzel participants.

Variable	Description	Type	Range	Notes
tas_q1	TAS-20 item 1 of 20, DIF (Difficulty Identifying Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q2	TAS-20 item 2 of 20, DDF (Difficulty Describing Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q3	TAS-20 item 3 of 20, DIF (Difficulty Identifying Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
tas_q4	TAS-20 item 4 of 20, DDF (Difficulty Describing Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. This item is reverse-keyed on the original scale; values in this column are already reverse-scored (i.e. summing tas_q1-tas_q20 directly reproduces tas , no further reversal needed). Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q5	TAS-20 item 5 of 20, EOT (Externally Oriented Thinking) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. This item is reverse-keyed on the original scale; values in this column are already reverse-scored (i.e. summing tas_q1-tas_q20 directly reproduces tas , no further reversal needed). Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q6	TAS-20 item 6 of 20, DIF (Difficulty Identifying Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
tas_q7	TAS-20 item 7 of 20, DIF (Difficulty Identifying Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q8	TAS-20 item 8 of 20, EOT (Externally Oriented Thinking) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q9	TAS-20 item 9 of 20, DIF (Difficulty Identifying Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q10	TAS-20 item 10 of 20, EOT (Externally Oriented Thinking) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. This item is reverse-keyed on the original scale; values in this column are already reverse-scored (i.e. summing tas_q1-tas_q20 directly reproduces tas , no further reversal needed). Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
tas_q11	TAS-20 item 11 of 20, DDF (Difficulty Describing Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q12	TAS-20 item 12 of 20, DDF (Difficulty Describing Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q13	TAS-20 item 13 of 20, DIF (Difficulty Identifying Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q14	TAS-20 item 14 of 20, DIF (Difficulty Identifying Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q15	TAS-20 item 15 of 20, EOT (Externally Oriented Thinking) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
tas_q16	TAS-20 item 16 of 20, EOT (Externally Oriented Thinking) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q17	TAS-20 item 17 of 20, DDF (Difficulty Describing Feelings) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q18	TAS-20 item 18 of 20, EOT (Externally Oriented Thinking) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. This item is reverse-keyed on the original scale; values in this column are already reverse-scored (i.e. summing tas_q1-tas_q20 directly reproduces tas , no further reversal needed). Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.

Variable	Description	Type	Range	Notes
tas_q19	TAS-20 item 19 of 20, EOT (Externally Oriented Thinking) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. This item is reverse-keyed on the original scale; values in this column are already reverse-scored (i.e. summing tas_q1-tas_q20 directly reproduces tas , no further reversal needed). Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.
tas_q20	TAS-20 item 20 of 20, EOT (Externally Oriented Thinking) subscale. Item wording not reproduced; see Bagby, Parker, & Taylor (1994).	integer	1-5	1 = strongly disagree; 5 = strongly agree. Missing for all 105 participants from the Monzel study, for whom only total/subscale scores were shared.